AI BASED INTERACTIVE ROBOT FOR ELDERLY PEOPLE

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Objective

- To make an AI based interactive robot for elderly people who are living alone.
- To Make a personal assistant to interact with them and to monitor their health.
- To Make a robot at low cost to make them affordable to all civilians.

Proposed Solution

AI based interactive robot can be designed which can interact with elderly at home, monitor their problems and raise alarms for family members. Interactive jovial talks can be relaxing and remedy for many ailments.

Project Novelty

- We make low cost interactive robot with good quality.
- The robot can interact with the elderly and monitor their problems.
- If any problems occur with their health it raises alarms to their family members or nearby hospital.

Literature survey

S.No	Authors	Title of the research work	Journal Name / Year	Inference
1	Ryosuke Miyata, Yunan He, Osamu Fukuda Nobuhiko Yamaguchi and Hiroshi Okumura, Anik Nur Handayani.	Vision-based Control for Open-source Mobile Robots	IEEE / 21 October 2020	How AI and Speech recognition is used in autonomous mobile robot. And how robot operating system (ROS) is used to give command to robot.
2	Zhiheng Wang, Perry Xiao, Daqing Chen.	Design of a Voice Control 6DoF Grasping Robotic arm Based on Ultrasonic Sensor, Computer Vision and Alexa Voice Assistance.	IEEE / 23-25 Aug. 2019	How Voice assistance is used in robotics and how it is interfaced.

Literature survey

S.No	Authors	Title of the research work	Journal Name / Year	Inference
3	Wuttichai Vijitkunsawat, Peerasak Chantngarm.	Comparison of Machine Learning Algorithm's on Self-Driving Car Navigation using Nvidia Jetson Nano	IEEE / 04 August 2020	How AI and Obstacle avoidance is used in autonomous mobile robot.
4	C. Park, S. Kang, J. Kim and J. Oh	A Study on Service Robot System for Elder Care	IEEE / 19 February 2013	How robots are used to serve for elderly care.
5	M. Aman, M. Mahmud, H. Jiang, A. Abdelgawad and K. Yelamarthi	A sensor fusion methodology for obstacle avoidance robot	IEEE / 08 August 2016	How obstacle avoidane is used for the movement of robot.

Literature survey

S.No	Authors	Title of the research work	Journal Name / Year	Inference
6	Chiao-Yu Yang, Ming-Jen Lu, Shih-Huan Tseng and Li-Chen Fu	A Companion Robot for Daily Care of Elders based on Homeostasis	IEEE / 13 November 2017	How robots are used in daily care for elders.
7	S. C. Mukhopadhyay and G. Sen Gupta	Sensors and Robotic Environment for Care of the Elderly	IEEE / 29 October 2007	What are the robotic environment for elderly people and he sensors.
8	Bhuvaneswari.P.T.V, Vignesh.S, Papitha.S, Dharmarajan.R.S	Humanoid Robot Based Physiotherapeutic Assistive Trainer for Elderly Health Care	IEEE / 26 June 2014	Humanoid robots assist for elderly care.

Literature survey

S.No	Authors	Title of the research work	Journal Name / Year	Inference
9	Yan Peng, Dong Qu , Yuxuan Zhong, Shaorong Xie, Jun Luo, Jason Gu.	The Obstacle Detection and Obstacle Avoidance Algorithm Based on 2-D Lidar	IEEE / 01 October 2015	Obstacle avoidance used to avoid collision with surroundings.
10	T R Madhavan, M Adharsh	Obstacle Detection and Obstacle Avoidance Algorithm based on 2-D RPLiDAR	IEEE / 02 September 2019	Obstacle detection and avoidance used to avoid collision with surroundings.

Methodology

The fast adoption and rising performance of voice platforms drive interest in smart homes. Home assistants can understand voice commands, whether for answering a question, playing music, or turning on the lights. To talk naturally to people, assistants should use a combination of AI technologies:

•**Automatic speech recognition** (ASR) is a voice processing component which converts spoken language into textual representation.

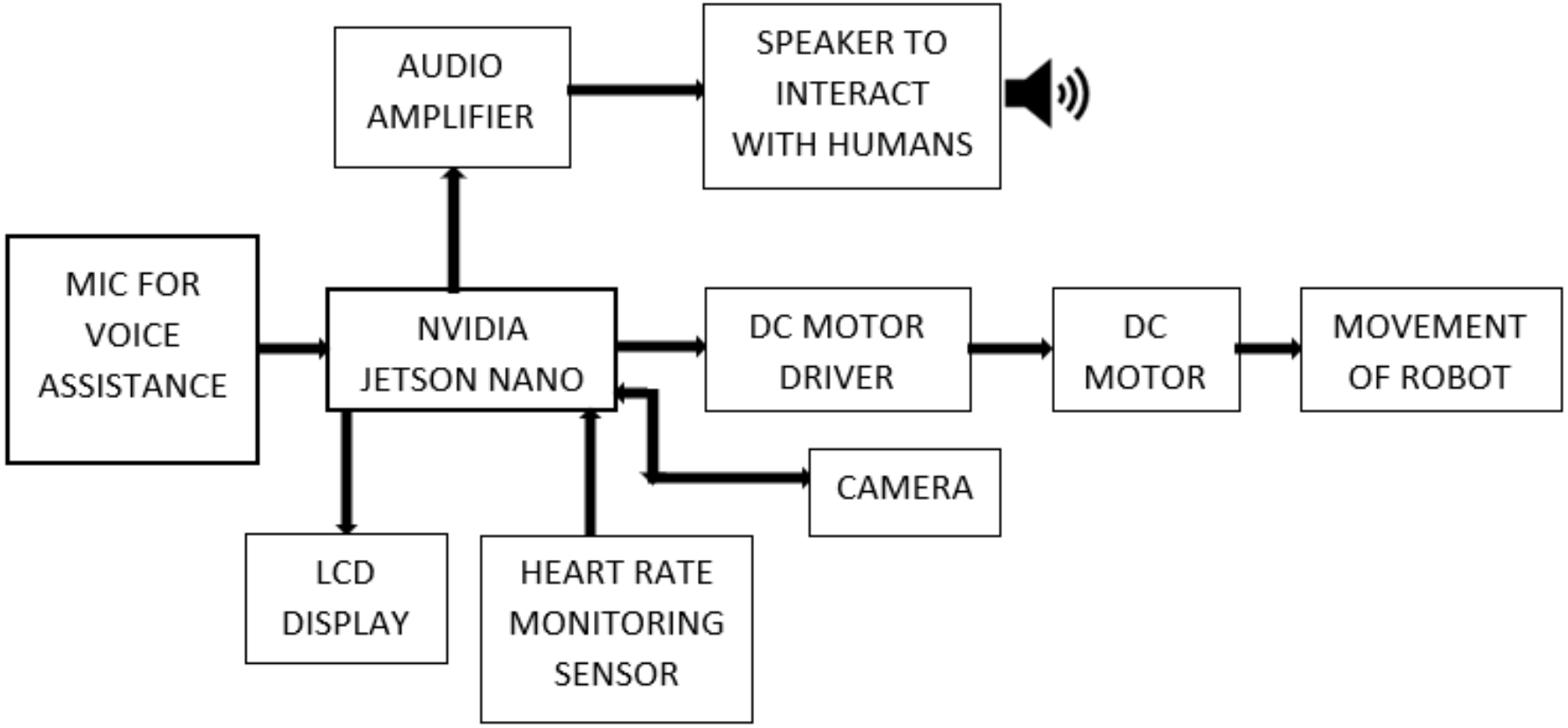
•**Natural language understanding** (NLU) translates user queries from natural language into a formal semantic representation. This representation is then used by the system to select the next action and generate a response. Usually, NLU requires completion of a variety of tasks, such as text normalization, morphological analysis, semantic analysis, ranking and named entity recognition.

•**Text-to-speech** (TTS) takes the response in a text format and produces its voice representation.

Methodology

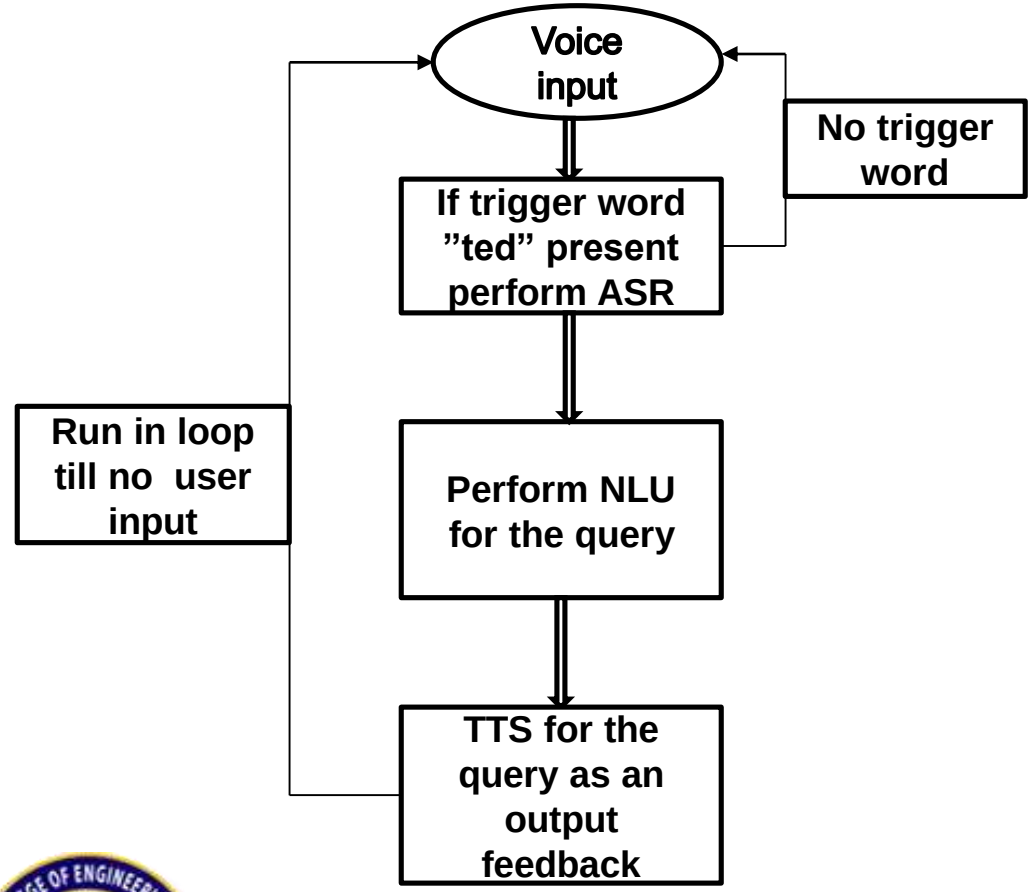
Obstacle avoidance robot is design to allow robot to navigate in unknown environment by avoiding collisions. Obstacle avoiding robot senses obstacles in the path, avoids it and resumes its running. We have make use of sensors to achieve this objective. The obstacle avoidance algorithm is the process or set of rules to be followed in calculating the data from the various sensors. The algorithm is a detailed step-by-step instruction set or formula for solving the problem of detecting all types of objects both moving or stationary.

Block Diagram

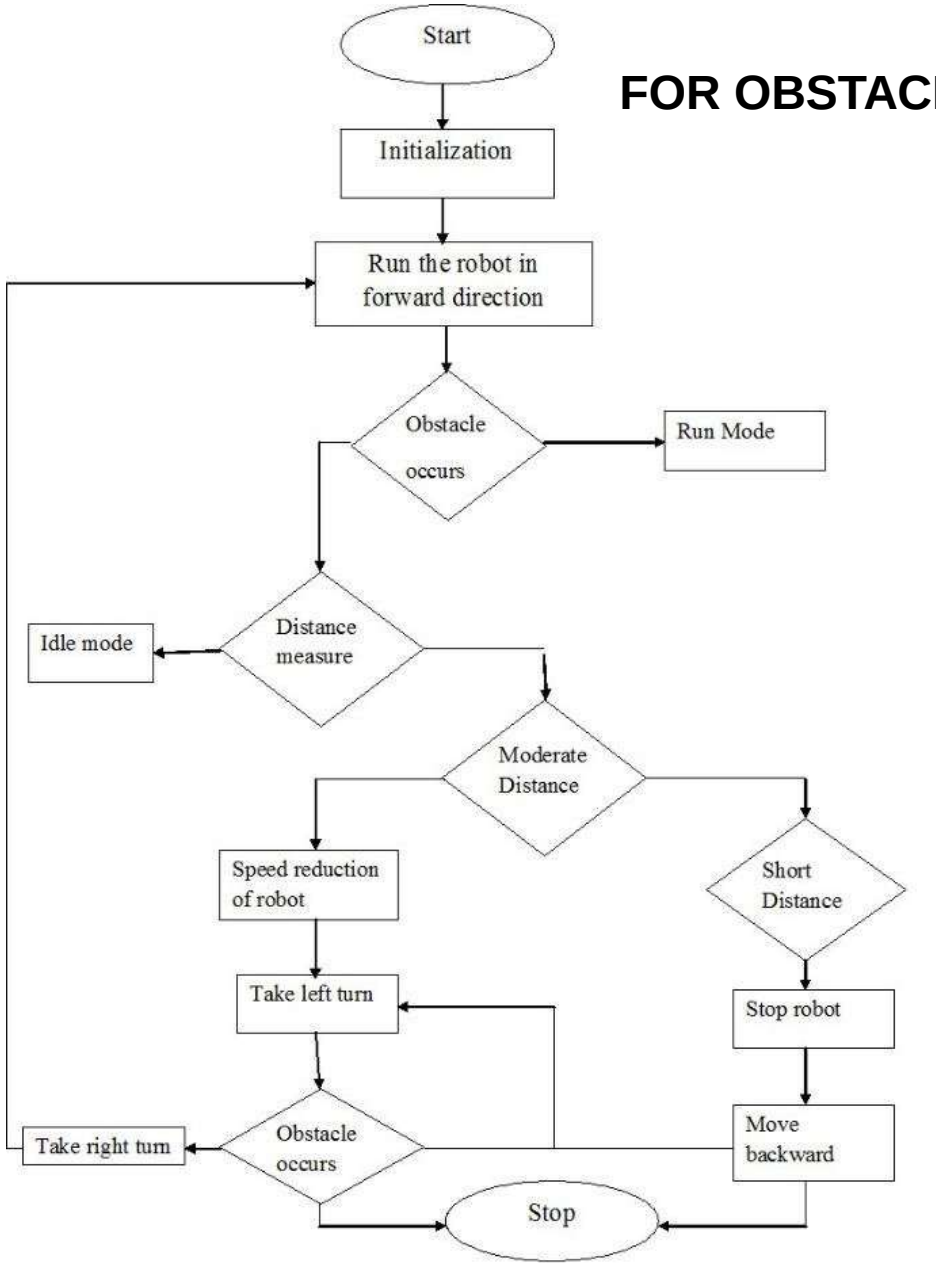


Flow Diagram

FOR VOICE ASSISTANT:



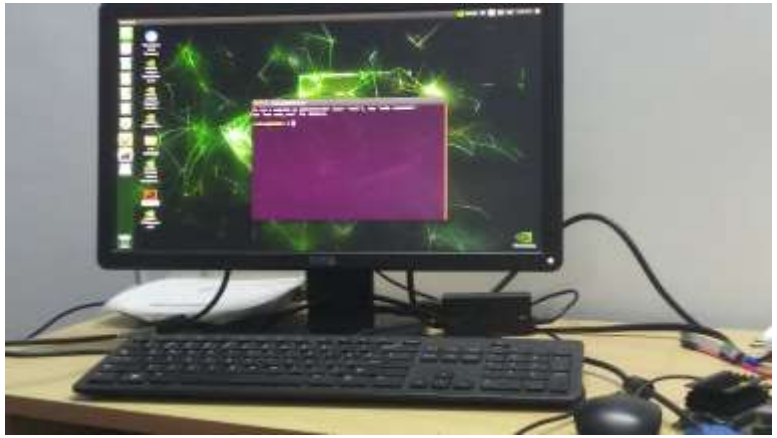
FOR OBSTACLE AVOIDANCE:



Tool used [Hardware / Software]

- NVIDIA JETSON NANO board (4gb)
- Camera & Mic
- LCD Display (5 inch)
- DC Motors (12V), 10kg high torque and DC Motor Driver
- Audio Amplifier and Speaker
- Heart Rate Monitoring sensor
- ROS (robotic operating system) used in Jetson nano

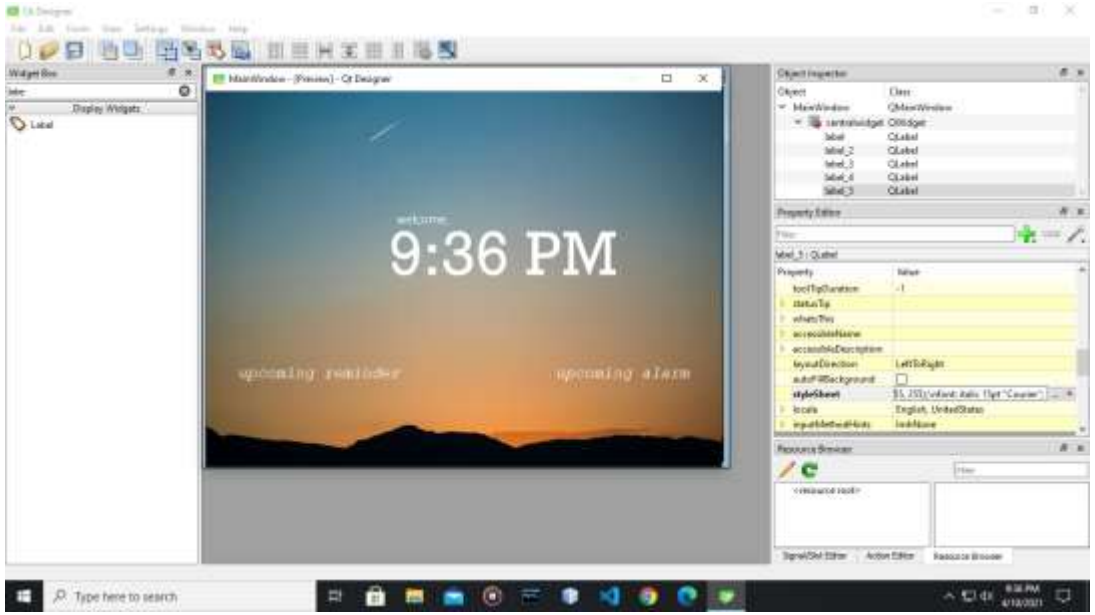
Work done & Results

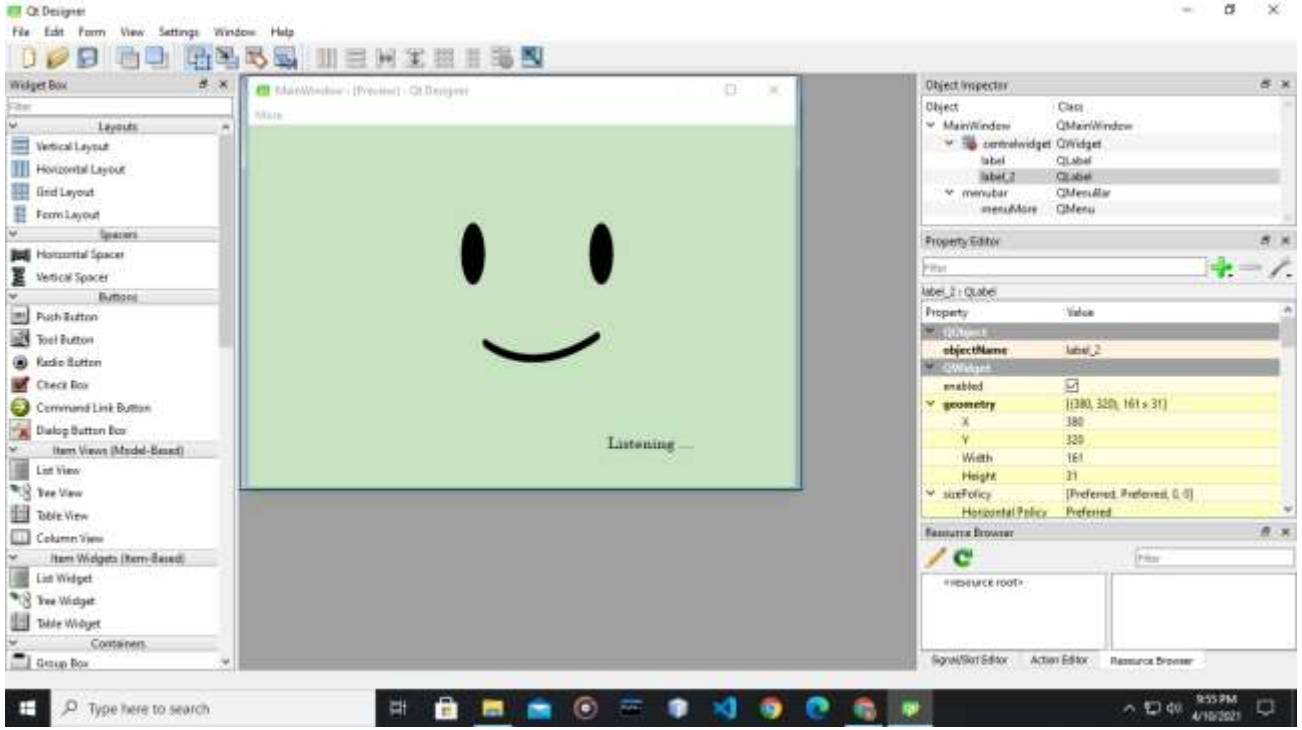


We have done all the structural design and created a voice assistant named “TED” and also obstacle avoidance for the movement of robot. It also has a GUI interface.

Work done & Results

GUI Application Development





Work Plan

Month / Week	Plan
Dec / (1-4)	Literature Survey / Zeroth Review / Project Approval
Jan / (1-2)	List of hardware components required
Jan / 3	Design of the structure of robot
Jan / 4	Specification of each components required
Feb / 1	Assembling the chassis
Feb / 2	Building the outer structure of the robot
Feb / 3	Assembling chassis and the outer structure
Feb / 4	Interfacing drivers for controlling dc motor
Mar / 1	Started to work on Jetson nano board
Mar / 2	Getting power supply for Jetson nano and dc motor driver
Mar / 3	1st Review
Mar / 20	New chassis for improving stability and voice assistant
Mar / 24	2nd Review
Mar / 28	Obstacle avoidance
Apr/ (7-14)	Thesis

Conclusion

Thus we have a AI-voice assistant along with Obstacle avoidance interfaced to the robot which can interact with the elderly people and takes care of them.

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Thank You